Bioinformatic approaches to regulatory genomics and epigenomics

376-1347-00L | week 07

Pierre-Luc Germain

ETH Zürich

Today’s plan

- Debriefing on the assignment
- DNA accessibility & ATAC-seq
- Practical
  - Part 1: exploration of ATAC-seq data
  - Part 2: differential motif accessibility analysis

Debriefing: Multiple motifs per peak

Q: Of all the peaks, what proportion contains a motif for the factor?

```{r}
moi_peaks <- matchMotifs(motif2, subject=peaks, genome=genome, out="positions")
moi_peaks <- moi_peaks[[1]]
length(moi_peaks)/length(peaks)
```

```
[1] 0.5133615
```

Debriefing: Multiple motifs per peak

Q: Of all the peaks, what proportion contains a motif for the factor?

```{r}
moi_peaks <- matchMotifs(motif2, subject=peaks, genome=genome, out="positions")
moi_peaks <- moi_peaks[[1]]
length(moi_peaks)/length(peaks)
```

```
[1] 0.5133615
```

Debriefing: Multiple motifs per peak

Q: Of all the peaks, what proportion contains a motif for the factor?

```{r}
moi_peaks <- matchMotifs(motif2, subject=peaks, genome=genome, out="positions")
moi_peaks <- moi_peaks[[1]]
length(moi_peaks)/length(peaks)
```

```
[1] 0.5133615
```

Several motif matches per peak are reported like this !

Debriefing: Multiple motifs per peak

Q: Of all the peaks, what proportion contains a motif for the factor?

```{r}
moi_peaks <- matchMotifs(motif2, subject=peaks, genome=genome, out="positions")
moi_peaks <- moi_peaks[[1]]
length(moi_peaks)/length(peaks)
```

```{r}
table(countOverlaps(peaks, moi_peaks))
```

```
    0    1    2    3    4    5 
 1898 1527  103   19    5    3 
```

Debriefing: Multiple motifs per peak

Q: Of all the peaks, what proportion contains a motif for the factor?

The correct way would be (not counting multiple motifs per peak):

```{r}
sum(overlapsAny(peaks, moi_peaks))/length(peaks)
```

```
[1] 0.4661041
```

Debriefing: Correct subject for motif matching

Q: Of all instances of that motif **in the genome** (or in one chromosome), what proportion is bound by the factor (i.e. has a peak)?

```{r}
moi_peaks <- matchMotifs(motif2, subject=peaks, genome=genome, out="positions")
moi_peaks <- moi_peaks[[1]]
sum(overlapsAny(moi_peaks, peaks))/length(moi_peaks)
```

```
[1] 1
```

Debriefing: Correct subject for motif matching

Q: Of all instances of that motif **in the genome** (or in one chromosome), what proportion is bound by the factor (i.e. has a peak)?

```{r}
moi_peaks <- matchMotifs(motif2, subject=peaks, genome=genome, out="positions")
moi_peaks <- moi_peaks[[1]]
sum(overlapsAny(moi_peaks, peaks))/length(moi_peaks)
```

```
[1] 1
```

```{r}
moi_chr1 <- matchMotifs(motif2, subject=chr1_coords, genome=genome, out="positions")
moi_chr1 <- moi_chr1[[1]]
sum(overlapsAny(moi_chr1, peaks))/length(moi_chr1)
```

```
[1] 0.002665357
```

Of all instances of that motif in the genome, what proportion is bound by the factor (i.e. has a peak)?

```
mmusculus <- import(genome, "2bit", which = as(seqinfo(genome), "GenomicRanges"))
motif_instances_genome <- findMotifInstances(mmusculus, motif, mc.cores=2)
```

```
## Note: motif [motif] has an empty nsites slot, using 100.
```

```
length(motif_instances_genome)
```

```
## [1] 6544422
```

```
motif_with_peaks = overlapsAny(motif_instances_genome, peaks)
sum(motif_with_peaks)
```

```
## [1] 16856
```

```
percentage2 <- sum(motif_with_peaks)/length(motif_instances_genome)*100
percentage2
```

```
## [1] 0.2575629
```

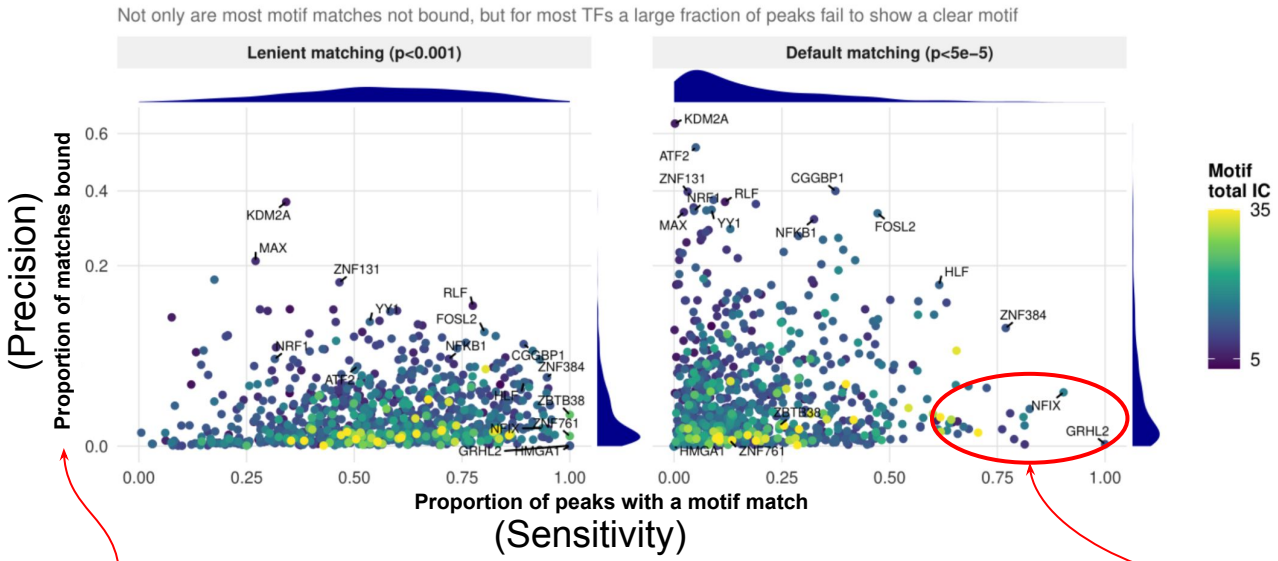
Of the 6544422 motif instances, 16856 (0.2575629%) overlap a peak.

TF: GATA1

Of the 9675 GATA1 peaks, 7277 (~75%) contain a GATA1 motif, but…

Debriefing on the assignment – wrapping up

Relationship between motif and ChIP-peaks across 976 TFs
(restricting the genome to regulatory elements)



Most motif matches are not bound

For a large fraction of factors, most binding sites don't show a strong motif

Those TFs for which most of the peaks have a motif also tend to have a very low fraction of the motif matches being bound → low-specificity motif

Example low-specificity motif (Mdx4) :

Example high-specificity motif (BCL6) :

DNA accessibility and ATAC-seq

DNA accessibility, which is associated to lower nucleosome density, **reflects activity**

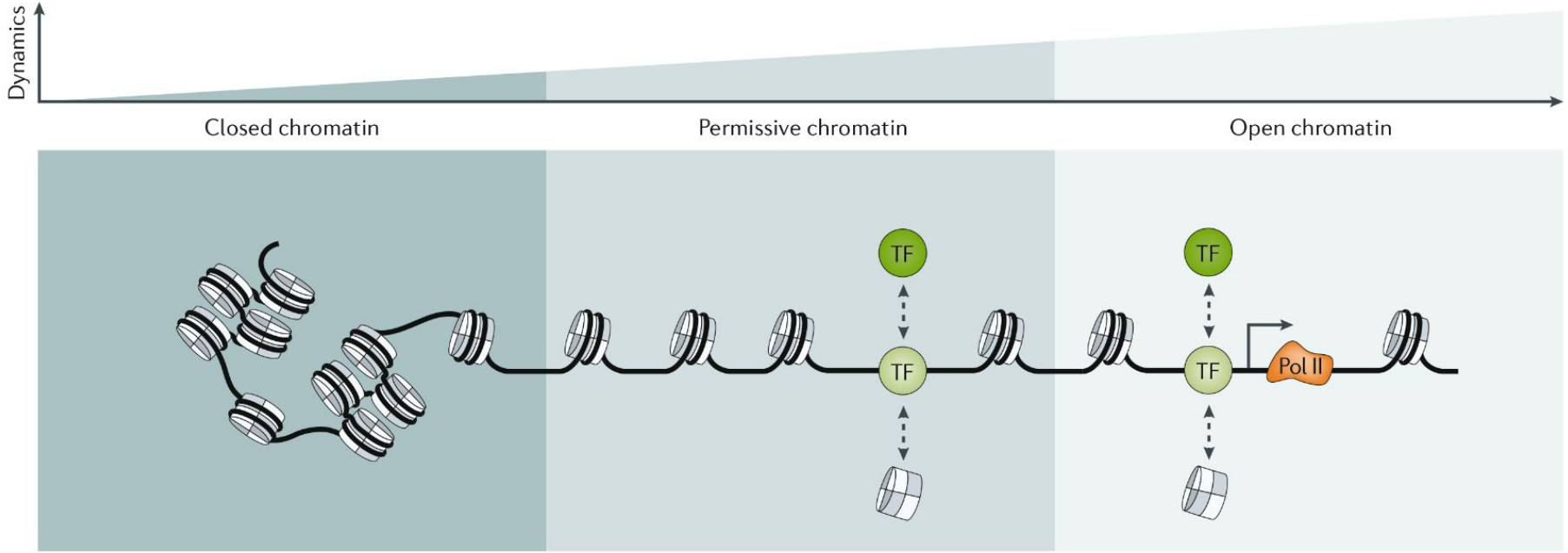
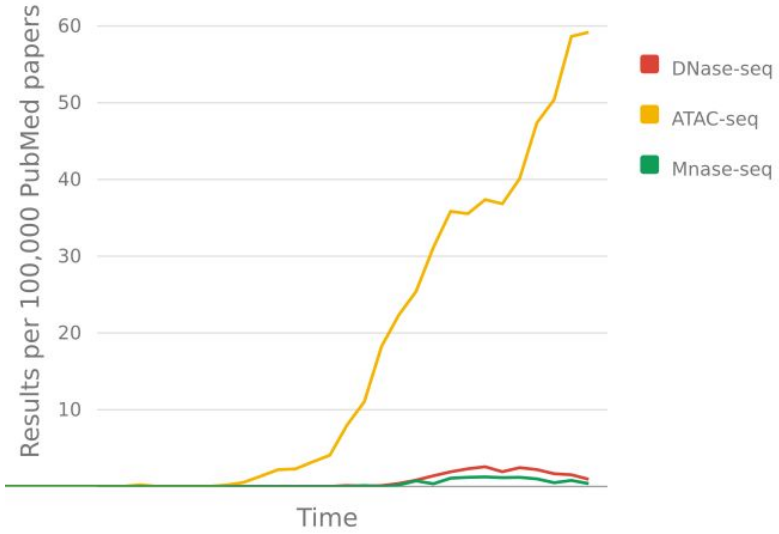


Fig. 1 | **A continuum of accessibility states broadly reflects the distribution of chromatin dynamics across the genome.** In contrast to closed chromatin, permissive chromatin is sufficiently dynamic for transcription factors to initiate sequence-specific accessibility remodelling and establish an open chromatin conformation (illustrated here for an active gene locus). Pol II, RNA polymerase II; TF, transcription factor.

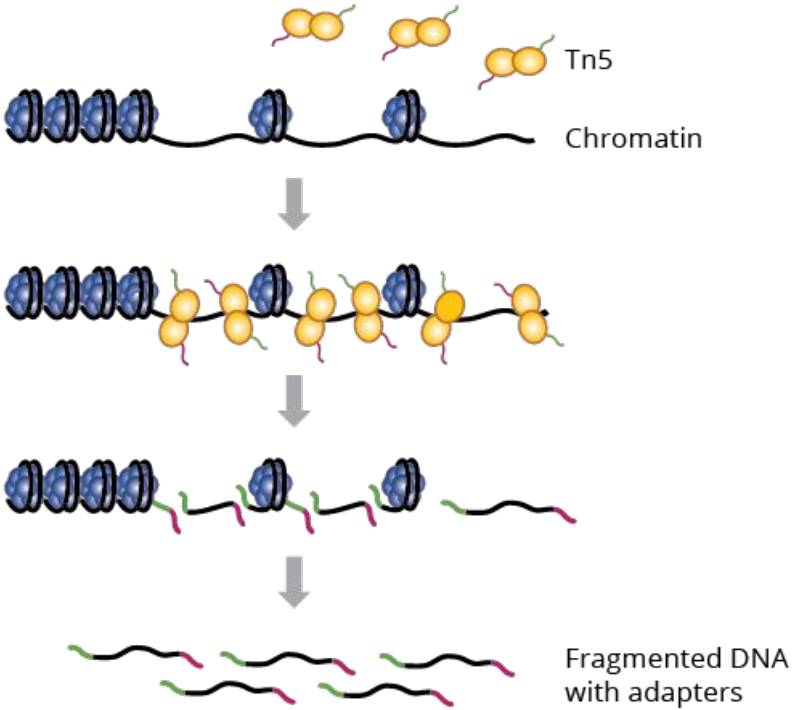
(Klemm, Shipony and Greenleaf, 2019)

Returning to our very brief history of genetics & genomics

…
1900 - Rediscovery of Mendel's work (1860s)
1913 - Chromosomes are linear arrays of genes
1941 - the one-gene-one-enzyme hypothesis
1944 - DNA is the genetic material
1951 - First protein sequenced
1977 - DNA sequencing
1977 - Eukaryotic genes are spliced
1995 - First bacterial genomes sequenced
2000 - Next Generation Sequencing (NGS)
2001 - Draft of the human genome
2003 - RNA-seq
2006 - ChIP-seq
2008 - DNAse-seq, MNase-seq
2012 - ATAC-seq
} Accessibility assays
…

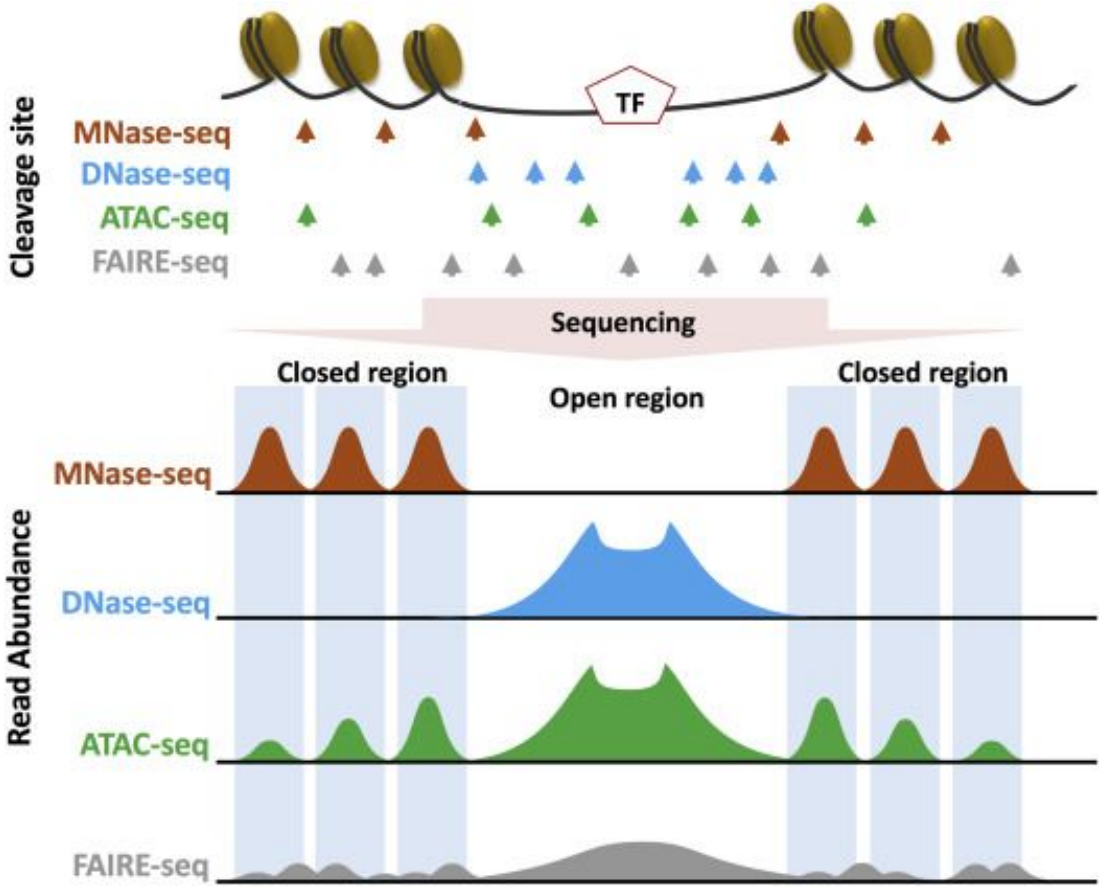


ATAC-seq



ATAC-seq recently became extremely popular due to its information content and low material requirement (i.e. # cells)

Chromatin accessibility assays

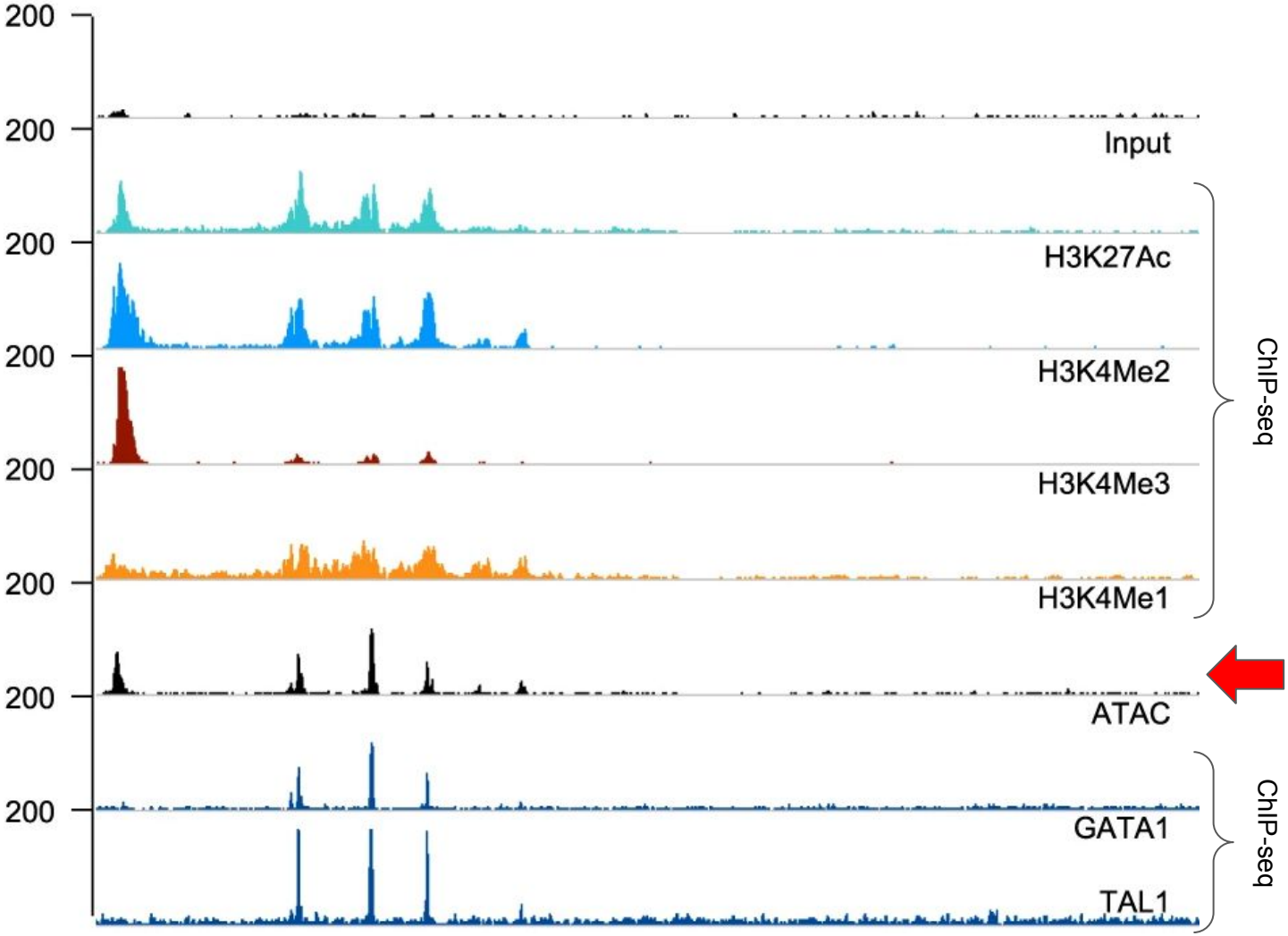


→ Nucleosome positions

→ DNase1-hypersensitive sites (DHS)
→ regulatory elements

Both

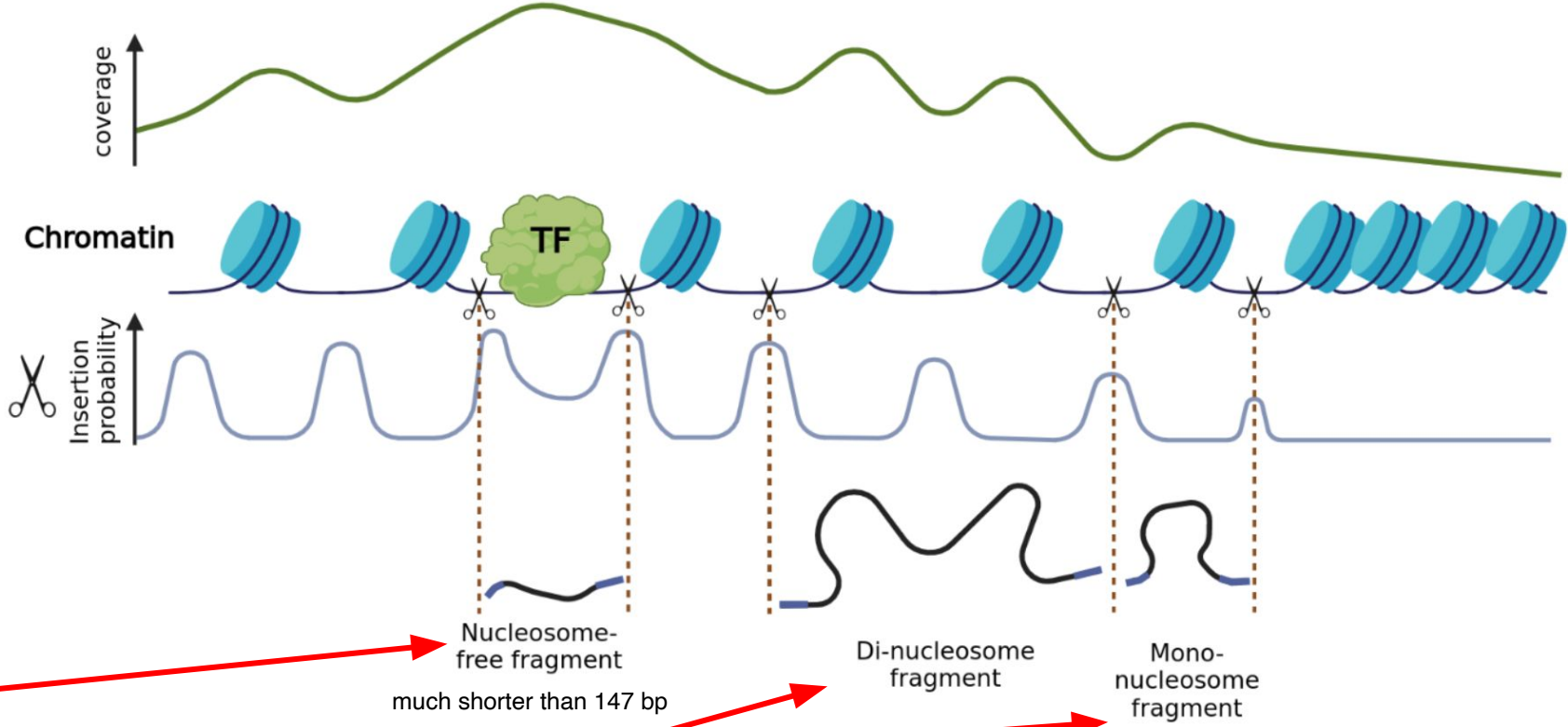
(Hsu et al., Epigenetics in Human Disease, 2018)



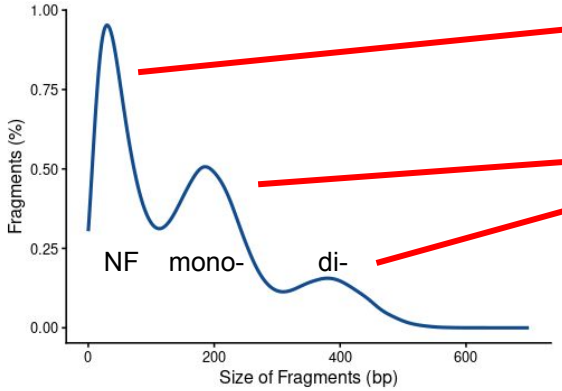
ATAC signal tells us that something is happening, it just doesn't tell us what exactly…

(Adapted from Fox et al., Nat Comm 2020)

ATACseq



much shorter than 147 bp

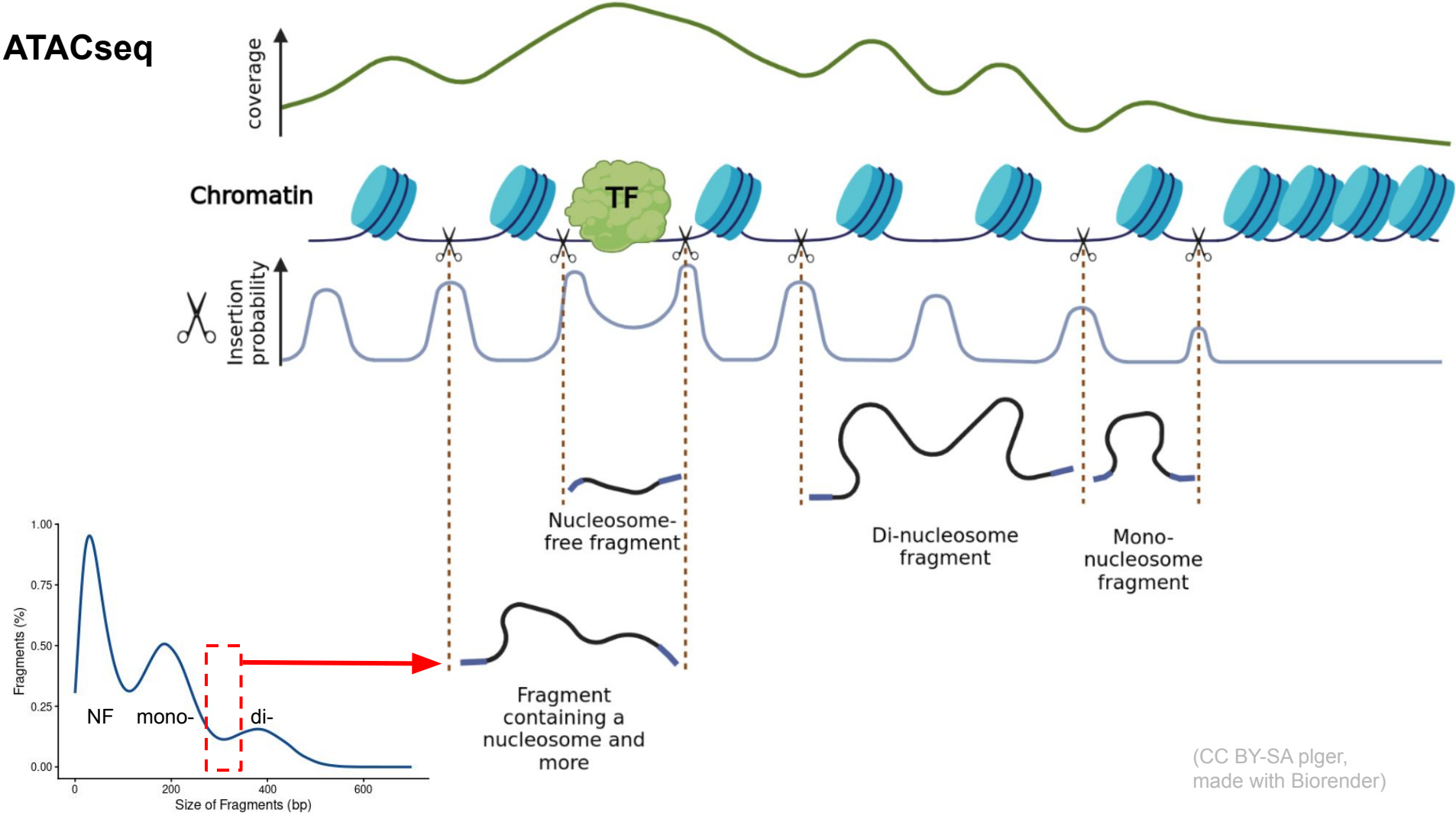


pure length of DNA wrapped around nucleosomes = 147bp
a lot of nucleosomes are protected with this H1 (linker histone) linker, generating longer fragments = around 190bp

how a good quality ATAC seq data should look like

(CC BY-SA plger, made with Biorender)

ATACseq

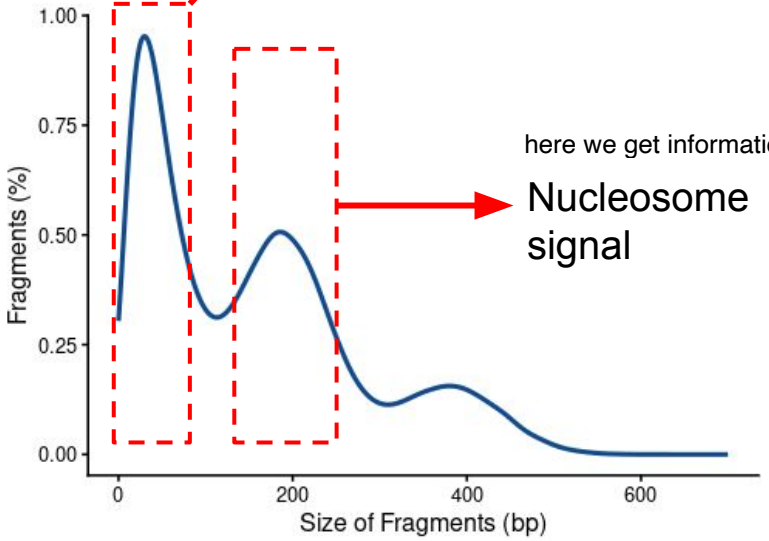


(CC BY-SA plger, made with Biorender)

This means that once we have the data, we can split the fragments according to size in order to obtain specific information about different kinds of chromatin signals

Nucleosome-free signal → Transcription factors

here we get information about TF binding and localization of regulatory sites



here we get information about where the nucleosomes are located

Nucleosome signal

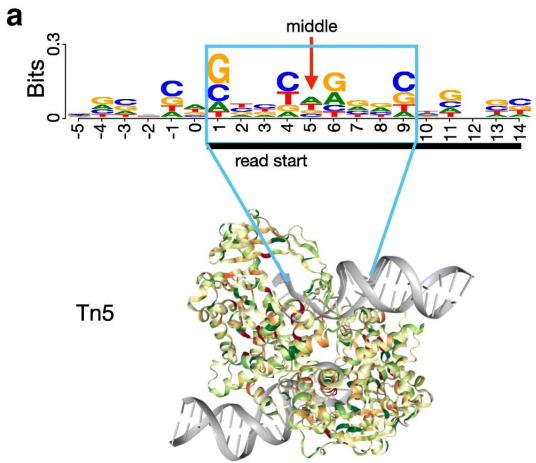
Practical - part 1

“Shifting” ATAC-seq alignments

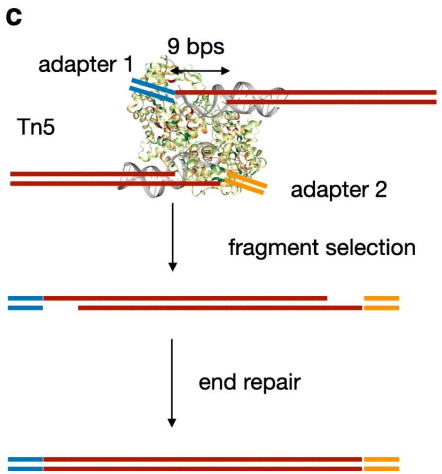
From a given ATAC-seq insertion site, the exact region that is accessible is a few nucleotides from the start of the read

When doing high-resolution things like footprinting, one therefore typically shifts the 5’ insertion site by +4/-5nt, so that it is placed in the middle of where the Tn5 was binding. (If using both ends, we can shift inwards by 4nt)

(For most other purposes, this is too fine-grained to make a difference)

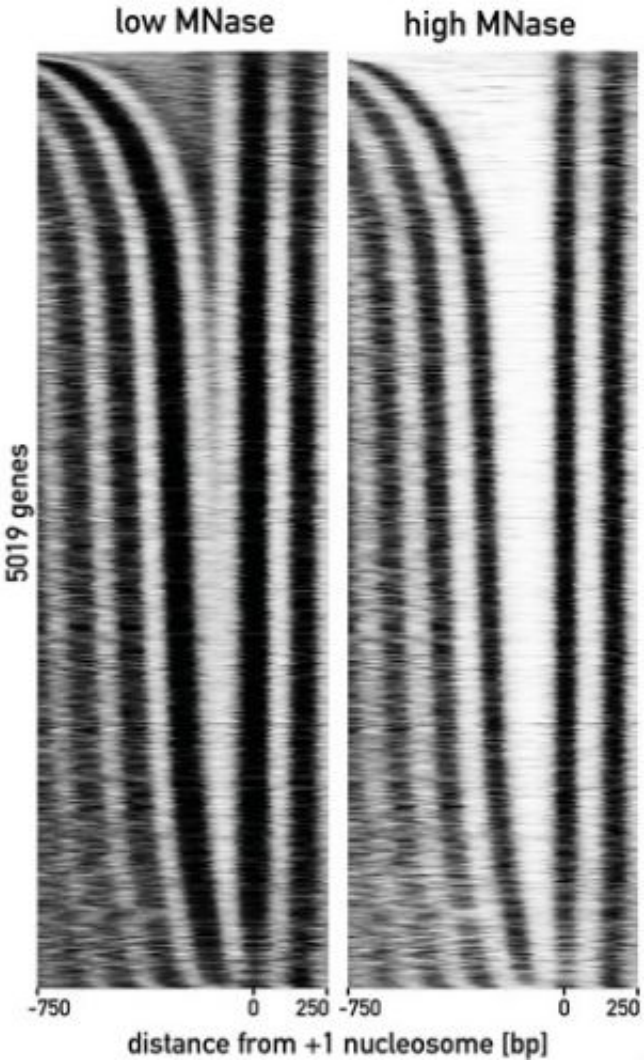


(adapted from Zhijian et al., Genome Biology 2019)



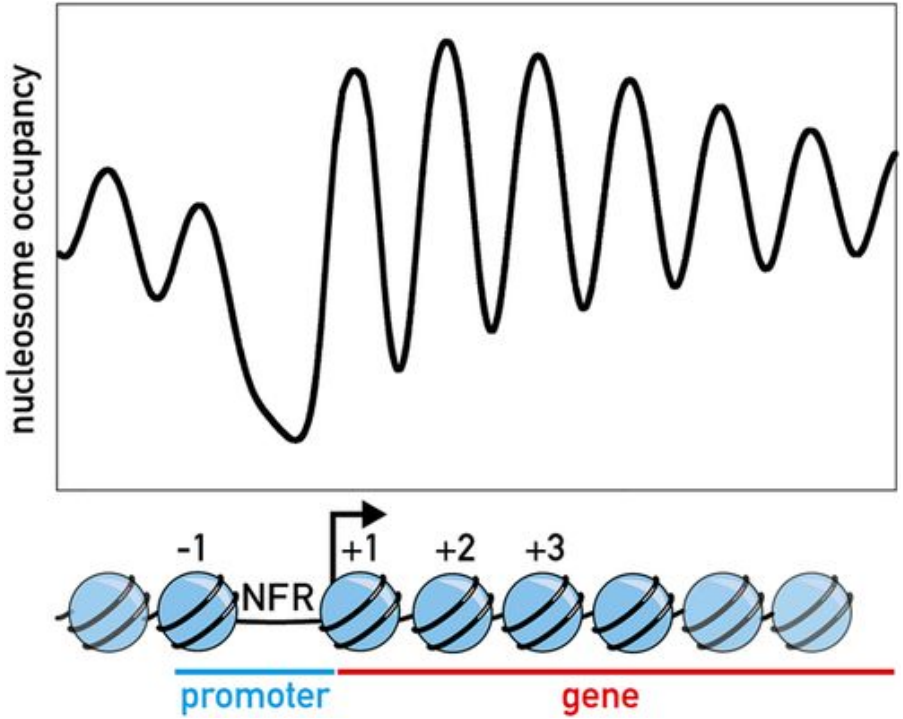
Nucleosome positioning

TF binding fixes neighboring nucleosome positions, and the phasing fades with distance

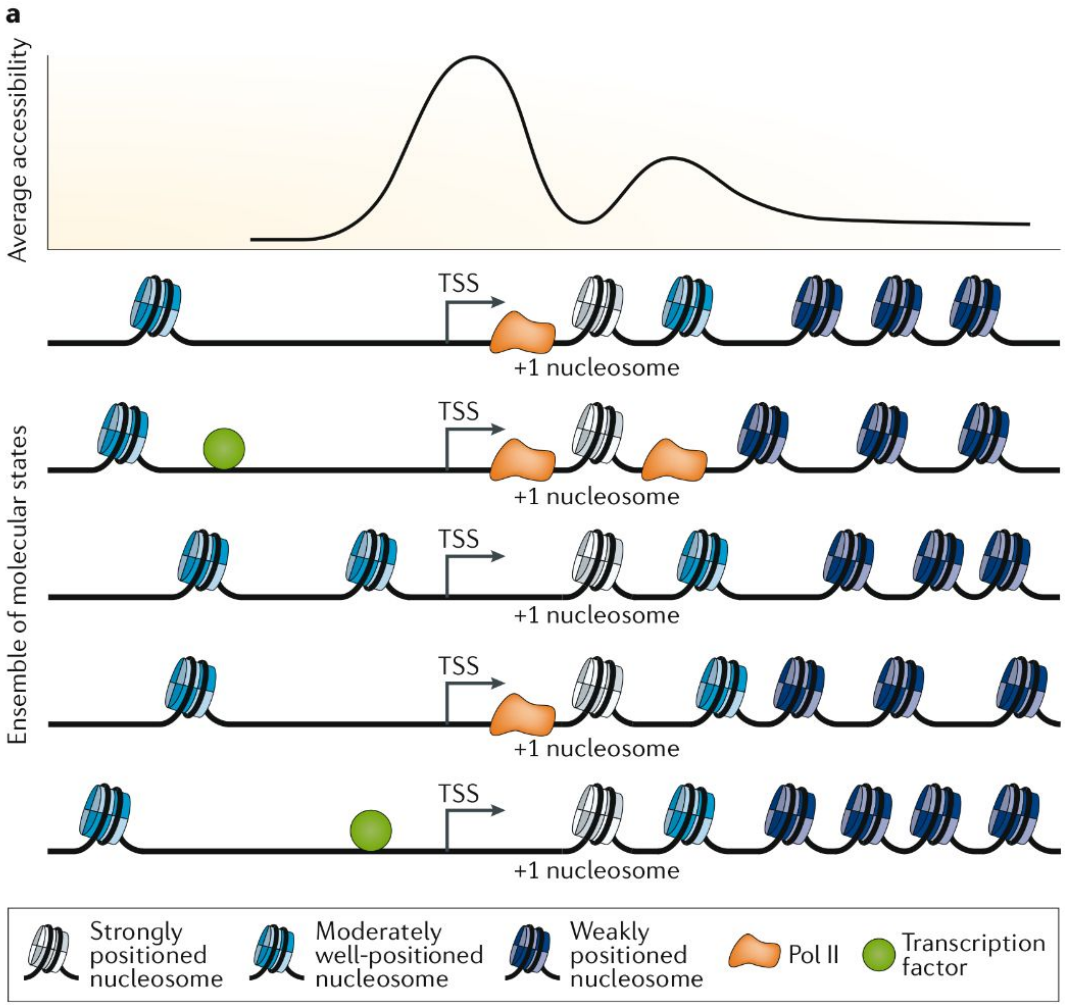


near the TSS nucleosomes are very well positioned

further away the TF are not there to restrict this movement and sliding of nucleosomes



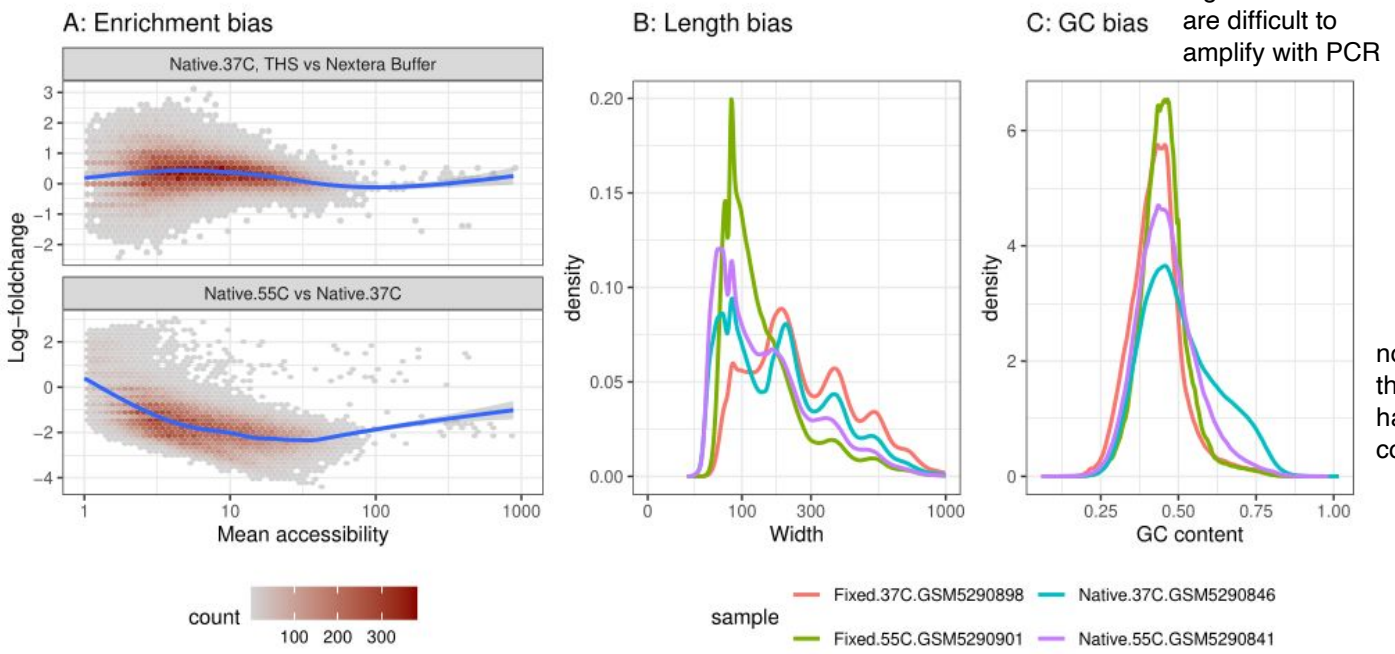
(Kubik, Bruzzone and Shore, 2017)



(Klemm, Shipony and Greenleaf, 2019)

3 (inter-related) types of technical sources of variation in ATAC-seq

high GC content are difficult to amplify with PCR



normally the genome has 0.5 CG content

these biass are not independent of each other:
regulatory elements are rich in GC
TSS are accessible, will therefore be shorter

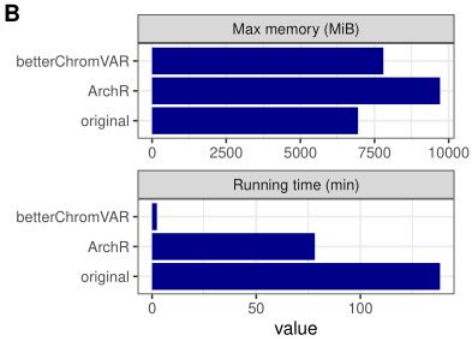
(Wang*, Sonder* et al., in preparation)

betterChromVAR enables correction of technical sources of variation and fast motif activity estimates

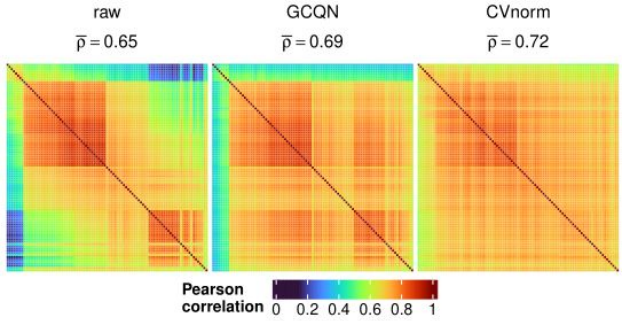
chromVAR is a very clever method to infer TF activity (i.e. overall motif accessibility), and it has been widely used in both bulk and single-cell data (Schep et al., *Nature Methods* 2017)

It explicitly corrects for enrichment and GC bias

We developed an analytical alternative (betterChromVAR) with full precision and massively increased scalability:



betterChromVAR::CVnorm also enables the normalization of technical sources of variations:



Pairwise correlation between ATAC-seq profiles of the same sample, performed with different reagents

(Wang*, Sonder* et al., in preparation)

Practical - part 2

Assignment

Part 1: In the same dataset of ATAC on chr19 used in the first part of the practical, plot

1) the insertion (i.e. ‘cuts’) profile of nucleosome-free fragments and

2) the centers of nucleosome-containing fragments,

around the high-confidence motifs of a transcription factor of your choice (one figure containing the two signals).

Part 2: In the peak counts dataset from stressed brains that we used in the second part of the practical, identify motifs that change in accessibility between males and females, and plot a heatmap of the deviations for the top few candidates motifs.

basically we do the same as in the second part of the practical but for male and female (instead of stress vs control)
if your looking for sex differences - include the stress as covariate (if you use all samples) - he actually forgot to implement sex as a covariate in his lm model

Please use section titles to delineate the two parts, and don't forget to render your markdown and push it as `assignment.html` !